LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 4: Loops and Polygons

Module 2 | 45 minutes | Counted loops (iteration) | Grades 6–8 · ages 11–14

Objective: Use counted loops to repeat instructions, and use the exterior-angle rule (360 ÷ sides) to draw any regular polygon.

Success criteria

- I can use a counted loop to remove repetition.
- I can calculate the turn for any polygon.
- I can use a variable and operator to generalise.

Key vocabulary

- **iteration**: Repeating a set of instructions.
- **counted loop**: A loop that repeats a fixed number of times.
- **exterior angle**: How far you turn at each corner.
- **generalise**: Make one solution work for many cases.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human polygon (5 min): a learner walks a square, triangle and hexagon in the hall; the class counts the total turning.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	The polygon machine: learners follow the steps.
14 min	**Perform**	Code Lab: hexagon, Pentagon turn, Code Lab: polygon machine
9 min	**Probe**	Loops in Python, activity
4 min	**Wrap-up**	Exit ticket: What is the turn angle for a 12-sided polygon?

Unplugged warm-up

Human polygon (5 min): a learner walks a square, triangle and hexagon in the hall; the class counts the total turning.

Teaching the idea

Counted loops implement iteration; with a variable and an operator, a single program generalises to any regular polygon (exterior angle 360/n). Highlights DRY code and the Python for-range equivalent.

Analogy: Walking around a building: you must turn a total of one full circle to face your starting direction again.

Live demo: Draw a square long-hand (8 blocks), then refactor into a loop; then into a variable polygon.

Questions to ask

Ask	Listen for
What do all the turns add up to?	360 degrees.
What is the turn for 10 sides?	36 degrees.
Why is the loop version better?	Shorter, clearer, easier to change.

Differentiation

Support
Provide a protractor sheet with 360/n worked out.

Stretch
Draw a star: repeat 5 { move 100; turn 144 }. Why 144?

Common misconceptions

Misconception	How to fix it
Turn by the inside angle	Turn by the exterior angle, 360 ÷ sides.
Loops only repeat move blocks	Any blocks can go inside.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What is the turn angle for a 12-sided polygon?

Homework
Find three regular polygons in the real world and work out each turn angle.

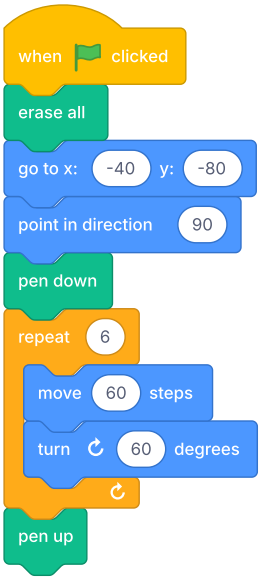
Answer key — Kitabash activities

1. Code Lab: hexagon (perform · code lab)

Goals checked: Closed shape with 6 sides · Uses a loop · Short code

One solution:

Pen



2. Pentagon turn (perform · multiple choice)

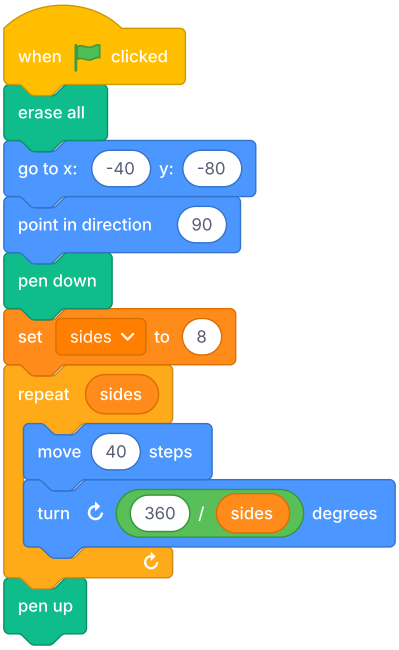
Answer: 72°

3. Code Lab: polygon machine (perform · code lab)

Goals checked: Draws a closed octagon · Calculates the turn · Uses the variable

One solution:

Pen



4. Loops in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: 360, turn, loop. **Example:** To get back to where it started facing the same way, the sprite has to turn a full circle, which is 360 degrees. The loop turns once per side, so each turn must be 360 divided by the number of sides.

Answer key — worksheet

1. **Key words:** iteration = Repeating a set of instructions. · counted loop = A loop that repeats a fixed number of times. · exterior angle = How far you turn at each corner. · generalise = Make one solution work for many cases.
2. **Plan your code:** Goals: Closed shape with 6 sides; Uses a loop; Short code.
3. **Pentagon turn:** A. 72° — 360 ÷ 5 = 72. (108° is the inside angle, not the turn.)
4. **Plan your code:** Goals: Draws a closed octagon; Calculates the turn; Uses the variable.
5. **Loops in Python:** A. hip hip hip hooray

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